DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 20: Scaled Limits

Finding the Right Scaling · The CLT · Donsker's Theorem

September 16, 2026

Today's Agenda

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3. The Central Limit Theorem: fixed-time limits

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2. The rescaled process
3. The Central Limit Theorem: fixed-time limits
4. Donsker's Theorem: the whole path
5. Tying back to Week 6

Learning Objectives

- Explain why $1/\sqrt{N}$ is the *only* scaling giving a non-trivial limit.

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- State how the CLT fixes the limiting distribution at each time.
- State Donsker's theorem and what it adds beyond the CLT.

Tuning a Microscope

Finding the Right Magnification

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Zoom horizontally only: the curve flattens to a boring line – structure destroyed. Zoom vertically too much: wiggles explode off-screen. Exactly *one* ratio keeps the picture interesting. For a random walk: compress time by N, space by $\sqrt{N}$.

Why $1/\sqrt{N}$ and Nothing Else

Rescale by $N^{-\alpha}$ and track the variance:

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The exponent isn't a choice – it's forced by the walk's own variance structure.

Definition

The Rescaled Random Walk

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Each step takes time $\Delta t = 1/N$ and has size $\pm 1/\sqrt{N}$.

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Space steps like the square root of time steps – the most important structural fact in stochastic calculus.

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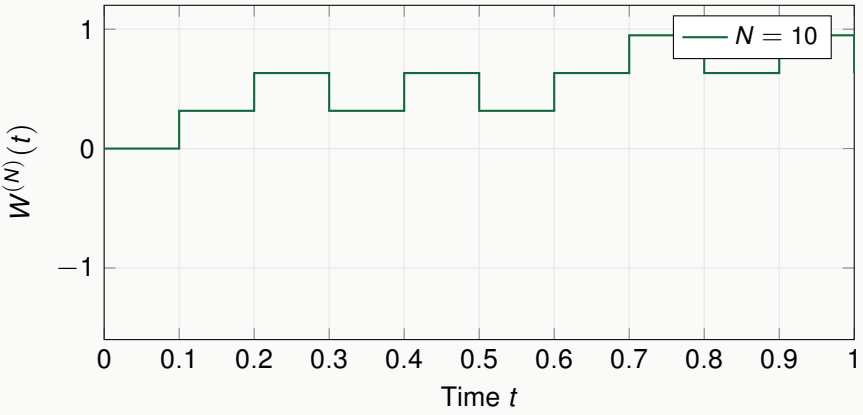
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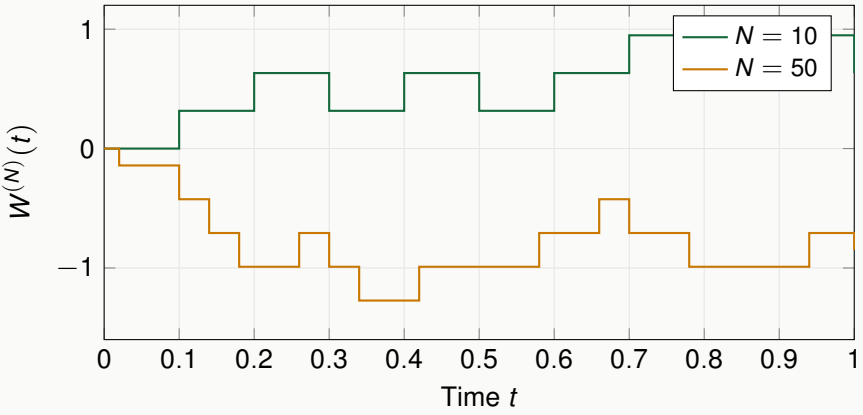
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Space steps like the square root of time steps – the most important structural fact in stochastic calculus. Exactly the CRR choice $u = e^{\sigma\sqrt{\Delta t}}$ from Lecture 16.

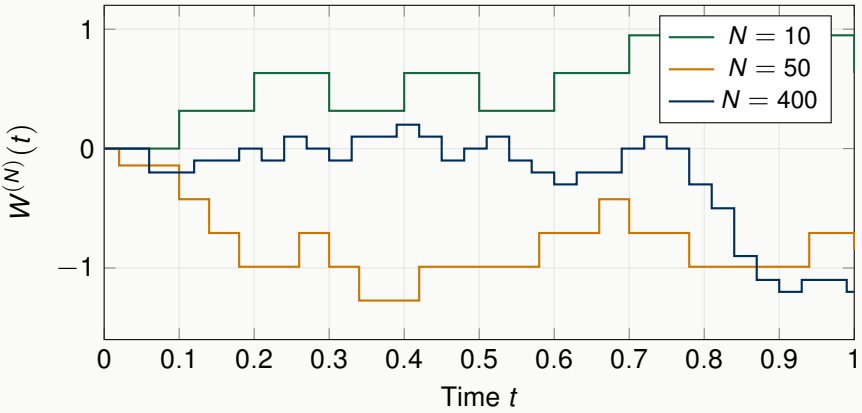
The Same Construction, Three Resolutions



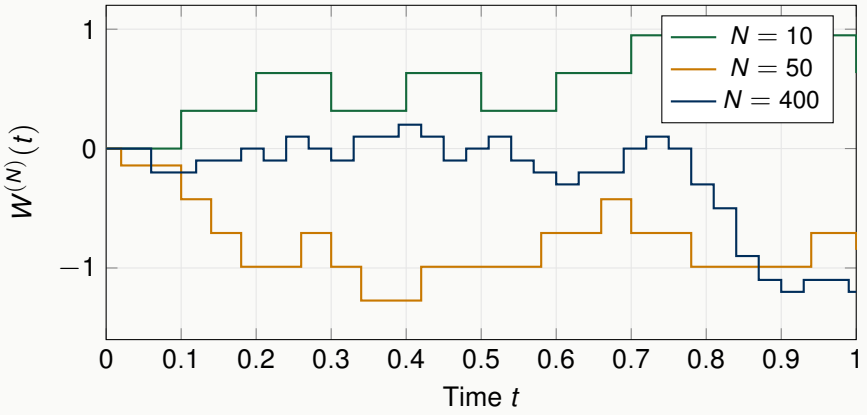
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Finer and rougher, but the *vertical extent* stays comparable – exactly what $1/\sqrt{N}$ achieves.

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Key Takeaway

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Where the variance t comes from

$\mathrm{Var}(S_{\lfloor Nt \rfloor}/\sqrt{N}) = \lfloor Nt \rfloor / N \to t$. **Variance grows linearly with time** – the hallmark of the limit process.

Simulated Convergence to Normality

100,000 rescaled walks, comparing $W^{(N)}(1)$ to $\mathcal{N}(0,1)$

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Variance is right by construction; the *shape* converges gradually. K–S halves each time N quadruples.

Donsker's Theorem

Functional Central Limit Theorem

The *entire* rescaled path $\{W^{(N)}(t)\}_{t\in[0,1]}$ converges in distribution, **as a random function**, to **standard Brownian motion** $\{W(t)\}$.

From Snapshots to the Whole Film

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Donsker doesn't need steps of exactly ± 1. *Any* i.i.d. steps with mean 0 and finite variance converge to the same Brownian motion. The fine details wash out entirely. This is why Brownian motion is such a ubiquitous assumption – you needn't believe prices move in literal ± 1 ticks.

The Week 6 Convergence Plot, Explained

Key Takeaway

Lecture 16 showed binomial prices converging to “the Black–Scholes price” without justification.

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Week 9 computes that limiting price in closed form.

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- **Donsker**: the whole path converges to standard Brownian motion.
- The limit is **universal** across step distributions.

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3. Why is the CLT alone insufficient to price a barrier option?

Looking Ahead

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Brownian Motion – Lecture 21

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Continuous everywhere. Differentiable nowhere.